

PHASE 2- Brainstorming & Ideation

Project Title: POCKETSMART AI

Your Smart Budget & Recommendation Assistant

Name: Government Arts and Science College, Manur.

College Code: MSU142

Team ID: 6ab21524d329d50f0db6120b

Team Leader: Esakki Aruna S **NM ID:**55055706AAD495BADB04C8C4D3B478B8

Team Member: Utchi Makali **NM ID:** 0BBEDB52C642EC0B0BA4CB4B22DE9E2D

Team Member: Kombu Madathi **NM ID:** 86EE18ABFB36AEF83784FA01234C5948

Team Member: S Ulagammal **NM ID:** B59DED29E07C4F78BCF442535560F984

1. Project Overview

PocketSmart AI is a smart personal budgeting and recommendation assistant designed to help users manage their money, understand spending patterns, set budgets, and make better purchase decisions. The system combines budgeting features with AI-based recommendations so that users can receive personalized suggestions based on their income, expenses, savings goals, and preferences.

2. Problem Statement

Many students and young adults find it difficult to track daily expenses, control unnecessary spending, and plan their monthly budgets. Traditional expense trackers mainly record transactions but may not explain where money is being spent or provide personalized suggestions. PocketSmart AI aims to address this gap by providing simple budget monitoring and intelligent recommendations in one application.

3. Main Idea

The core idea is to create an easy-to-use AI assistant that acts like a digital budgeting companion. Users can enter their income and expenses, define financial goals, and receive recommendations such as where they can reduce spending, how much they can allocate to different categories, and whether a planned purchase fits within their budget.

4. Objectives

- Track income and expenses in an organized manner.
- Create personalized monthly and category-wise budgets.
- Identify spending patterns and unnecessary expenses.
- Provide AI-based recommendations according to the user's budget and goals.
- Help users plan savings and monitor progress toward financial goals.
- Provide simple alerts when spending approaches or exceeds a budget limit.
- Present financial information using clear summaries and charts.

5. Target Users

- College students managing limited monthly allowances.
- Young adults starting to manage personal finances.
- Users who want a simple alternative to complicated budgeting tools.
- People who want recommendations before making non-essential purchases.

6. Brainstormed Features

Expense Tracking: Record expenses with categories such as food, travel, education, shopping, and entertainment.

Income Management: Record salary, allowance, part-time income, or other sources of money.

Smart Budget Creation: Suggest category budgets using the user's income, spending history, and savings target.

AI Recommendation Assistant: Answer budget-related questions and provide personalized suggestions.

Spending Analysis: Identify high-spending categories and recurring spending patterns.

Purchase Decision Support: Allow users to enter a planned purchase and receive a budget-based recommendation.

Savings Goal: Set a target amount and track progress toward it.

Budget Alerts: Notify users when category spending is close to the selected limit.

Dashboard: Display income, expenses, remaining budget, savings, and spending trends.

Monthly Report: Generate a simple summary of the user's monthly financial activity.

7. AI Recommendation Ideas

- Suggest ways to reduce spending in categories where the user is consistently exceeding the budget.
- Recommend an affordable spending limit for a planned purchase.
- Suggest a savings amount based on income and regular expenses.
- Compare current spending with previous months and explain major changes.
- Provide reminders and simple tips based on the user's financial goals.
- Answer natural-language questions such as: “Can I afford this purchase this month?”

8. Proposed Workflow

1. User creates an account and enters basic financial information.
2. User records income, expenses, budgets, and savings goals.
3. The system organizes the data into categories.
4. The AI analyzes spending patterns and budget status.
5. The system generates personalized recommendations and alerts.
6. The dashboard displays the user's current financial position and progress.

9. Possible Technology Stack

- Frontend: HTML, CSS, JavaScript or a suitable mobile/web UI framework.
- Backend: Python with Flask or Django.
- Database: MySQL, SQLite, or another suitable database.

- AI/Recommendation Layer: Python-based recommendation logic and, if available, an AI/LLM service.
- Visualization: Chart.js or a Python visualization library for reports and graphs.

10. Innovation / Unique Value

The proposed system combines expense tracking, budgeting, savings goals, and personalized recommendations rather than functioning only as a digital expense diary. Its main value is converting recorded financial data into understandable suggestions that can help users make informed everyday spending decisions.

11. Expected Outcome

The final project is expected to provide a simple and interactive budgeting assistant that helps users understand their spending, stay within planned budgets, work toward savings goals, and receive personalized recommendations through an AI-assisted interface.

12. Future Enhancements

- Bank or payment transaction integration where legally and technically appropriate.
- Voice-based interaction with the AI assistant.
- Receipt scanning and automatic expense categorization.
- Personalized financial insights using longer-term spending history.
- Multi-language support, including Tamil and English.
- Secure cloud synchronization across devices.

13. Brainstorming Summary

PocketSmart AI is envisioned as a practical and student-friendly smart budgeting application. The brainstorming phase focuses on combining financial tracking with AI-assisted recommendations, while keeping the interface simple, understandable, and useful for everyday money management .